Bee Mortality Sptial Analysis from Bijensterfte Survey Data 2024

Presented by: Ming Zhang
9.12.2025

198

Responses

Beefoundation

meters maken voor bijen, hommels en vlinders

Section 1 of 8

Honingbijen sterfte 24-25

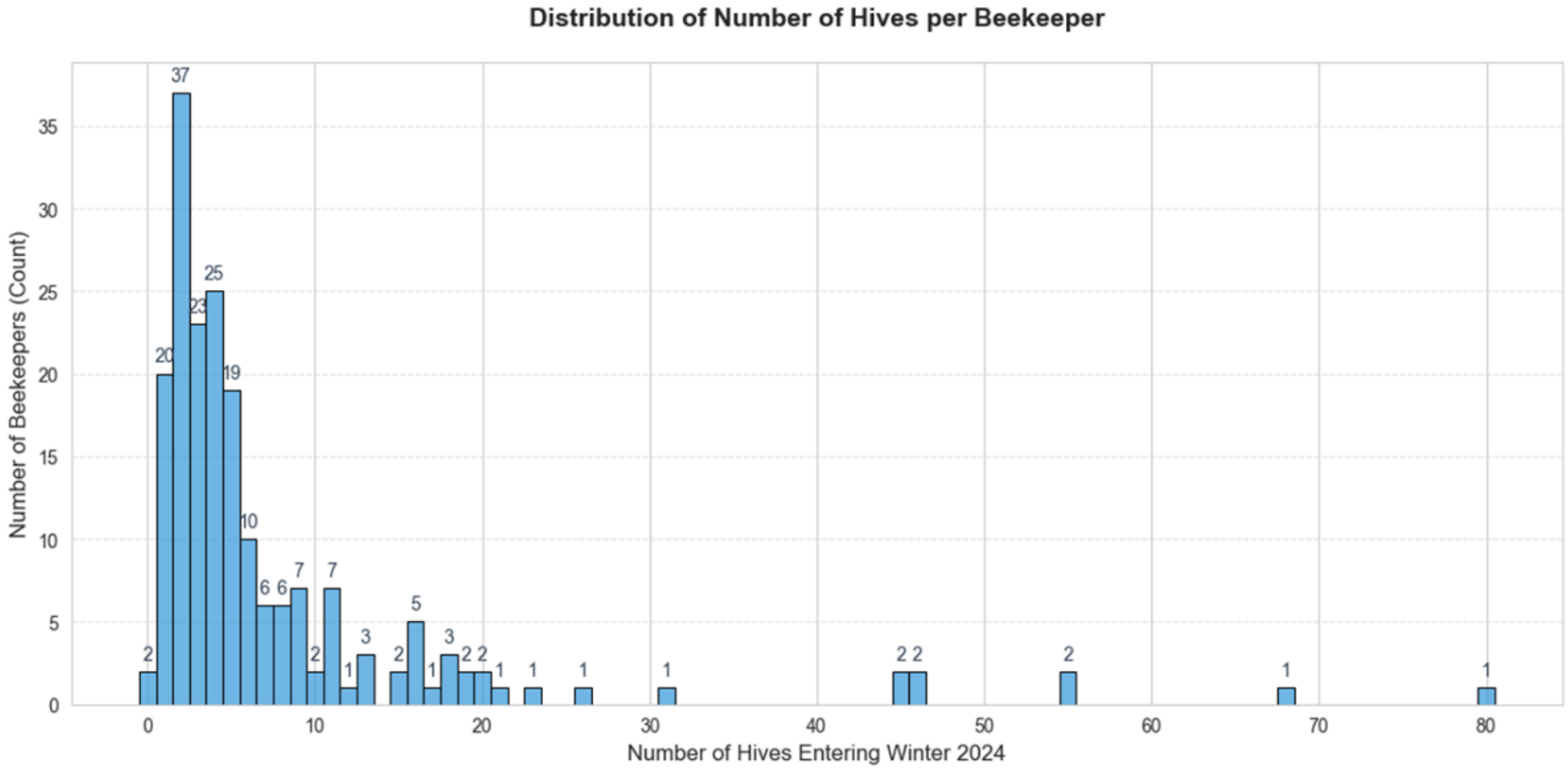
Beste imker,
Voor het derde jaar horen we van imkers dat de afgelopen zomer, herfst en winter weer veel bijenvolken gestorven zijn. Dit verlies is voor ons allen zeer pijnlijk. Bovendien: honingbijen zijn een wezenlijk deel van onze leefomgeving. Zij spiegelen de gezondheid van de omgeving waarvan wij allemaal deel uitmaken. Wat is er de afgelopen jaren aan het veranderen waardoor de sterfte zulke grote proporties aanneemt?

Waarom deze enquête?
Met deze enquête proberen we de vinger achter dit sterven te krijgen. Is het blootstelling aan pesticiden, klimaatverandering, de bedrijfsmethode van de imker, ziekteverwekkers of voedselaanbod? Of is het een combinatie van factoren? En het allerbelangrijkste: wat kunnen we doen om deze sterfte te keren?

Er lijkt een grillig sterftepatroon te zijn: op de ene plek zijn alle volken dood, op de andere is er niets aan de hand. Dit maakt dat het belangrijk is om gegevens te verzamelen van veel verschillende imkers dicht bij elkaar in de buurt. Deel daarom deze enquête met uw buurimkers. Als we samen meer gegevens verzamelen, krijgen we samen een beter inzicht.

Descriptive View of the Dataset

- The distribution of number of hives among respondents is highly right-skewed. The vast majority of the study population consists of small-scale, hobbyist beekeepers.
- At the beginning of the 2024 winter season, over 80% of respondents owns 10 or fewer hives. Therefore, our data to a large extent represents the average beekeeper's experience.



Professional
9
13.7%

Hobbyist
189
25.7%

Mortality Rate

Data Conversion: from beekeeper to hives

- The original survey contained beekeeper-level records, where one entry may include multiple hives. To analyze the possible effects at the hive level, this dataset was converted into hive-level entries (where applicable).

	A	B	C	D	E	F	G	H
1	**id**	**Postcode cijfers**	**Postcode Letters**	**GPS_x**	**GPS_y**	**Province**	**2024_spring**	**2024_winter**
2	id	Postcode cijfers	Postcode Letters	Vul hier je GPS locatie in		In welke provincie staan je bijen?	Met hoeveel volken begon je dit bijenjaar (voorjaar 2024?)	Hoeveel volken heb je ingewinterd? (najaar 2024)
4	1	10XX	XX			Utrecht	8	12
5	2	73XX	XX			Gelderland	2	2
6	3			52.XXXXX	6.XXXXX	Overijssel	3	4
7	4	30XX	XX			Zuid-Holland	4	4
8	5	70XX	XX			Gelderland	6	4
9	6			52.XXXXX	5.XXXXX	Noord-Holland	1	2

Beekeeper-level entries from the survey

	A	B	G	H	I	J	K	L	N	O
1	hive_id	hive_lifecycle_status	id	Postcode cijfe	Postcoc	GPS_y	GPS_x	Province	Death_rate_winte	Growth_rate_24_25
2	2_hive_1	Survived Winter 24	2	73XX	XX			Gelderland	0.5	-0.
3	2_hive_2	Died During Winter 24	2	73XX	XX			Gelderland	0.5	-0.
4	3_hive_1	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
5	3_hive_2	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
6	3_hive_3	Survived Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
7	3_hive_4	Died During Winter 24	3			52.XXXXX	6.XXXXX	Overijssel	0.25	
8	4_hive_1	Survived Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
9	4_hive_2	Survived Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
10	4_hive_3	Died During Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
11	4_hive_4	Died During Winter 24	4	31XX	XX			Zuid-Holland	0.5	-0.
12	5_hive_1	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
13	5_hive_2	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
14	5_hive_3	Survived Winter 24	5	73XX	XX			Gelderland	0.25	-0.
15	5_hive_4	Died During Winter 24	5	73XX	XX			Gelderland	0.25	-0.
16	5_hive_5	Lost Before Winter 24	5	73XX	XX			Gelderland	0.25	-0.

Hive-level entries

Data Conversion

- The first step is to identify the total number of hives each beekeepers have had.
  - Identify the number of hives that belongs to each beekeeper by identifying the largest number from *2024_spring*, *2024_winter*, and *2025_spring*. If a beekeeper reported n hives at a location, create n rows in Excel, each representing one hive.
- Add a few columns to describe the attributes of each hives:
  - *hive_id*: the unique identifier of a hive. In total, there are 2566 unique hives.
  - *hive_lifecycle_status*: whether the hive is lost before winter 2024, died during winter 2024, or survived winter 2024.

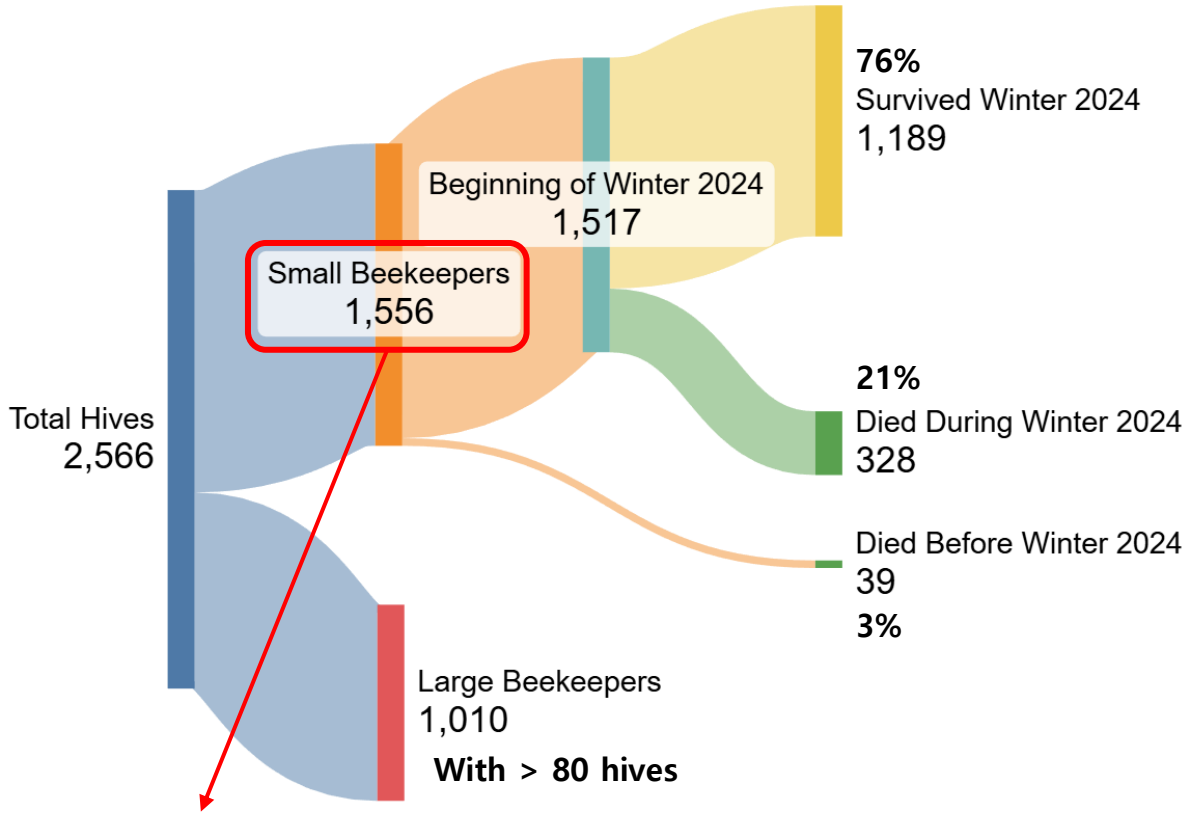
Beekeeper -level entries from the survey



Hive-level entries

Descriptive View of the Dataset

- The data cleansing process found that one beekeeper (id = 1) didn't give the number of hives in 2025 spring, and two beekeepers (id = 23 & 51) has a growth of hives during the winter. These three beekeepers are omitted from the dataset immediately.
- The dataset of the beehives continues with the data from 195 beekeepers. Altogether, they reported 2566 hives.
- There are two beekeepers who possess a substantial number of hives: id = 108 starts with 500 hives in spring 2024, and id = 120 starts with 80 hives in spring 2024.



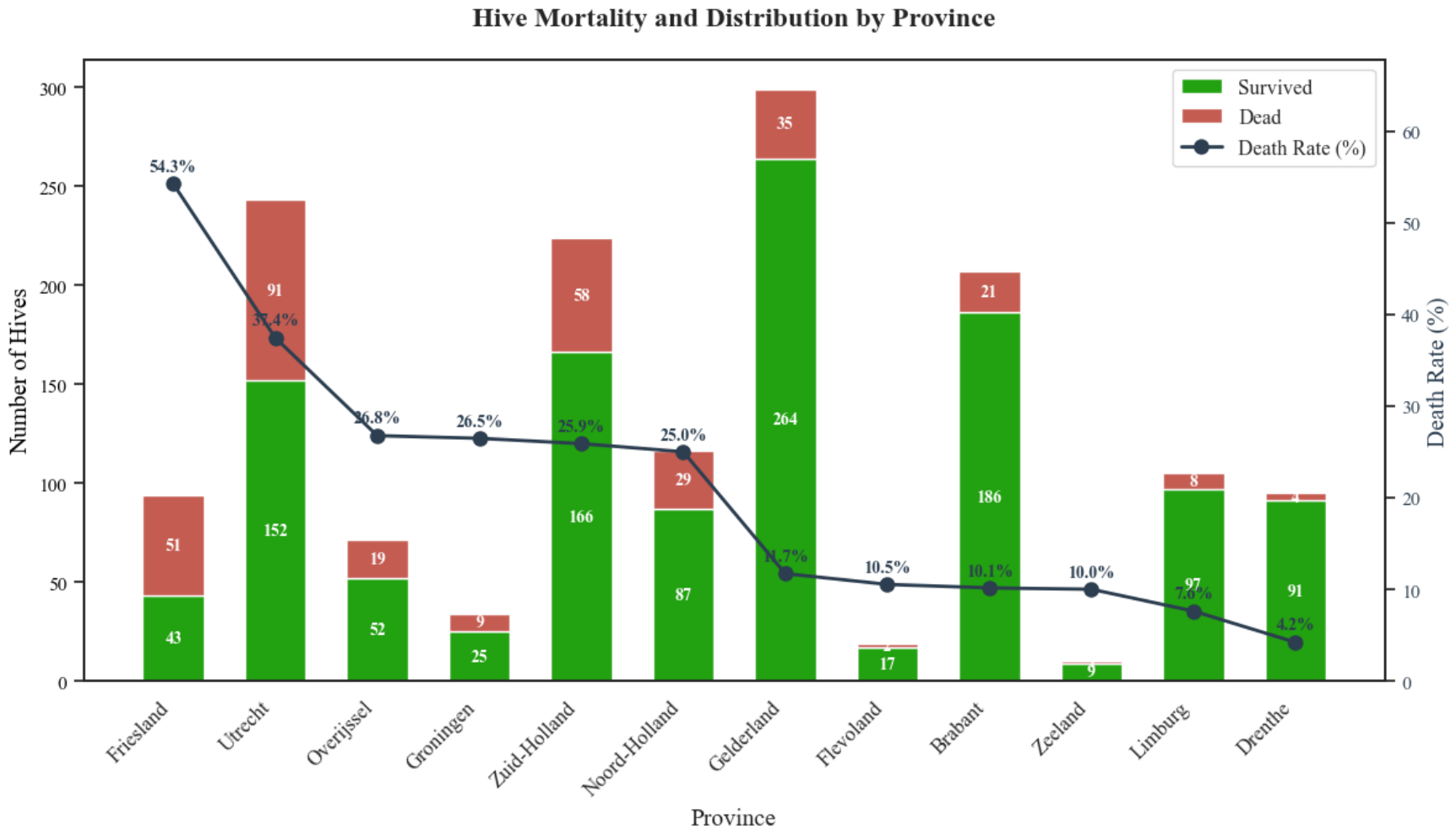
The following analysis focuses on the 1556 hives from small beekeepers.

24%

Mortality Rate

Descriptive View of the Dataset

- While mortality rates show high variance across provinces, a previous study by the Bee Foundation indicates that results are likely representative only for provinces with more than 300 hives.
- In this survey, while no single province met the 300-hive benchmark for robust statistical inference, this visualization still reveals province-level heterogeneity and serves as a valuable exploratory tool for identifying potential high-risk regions.

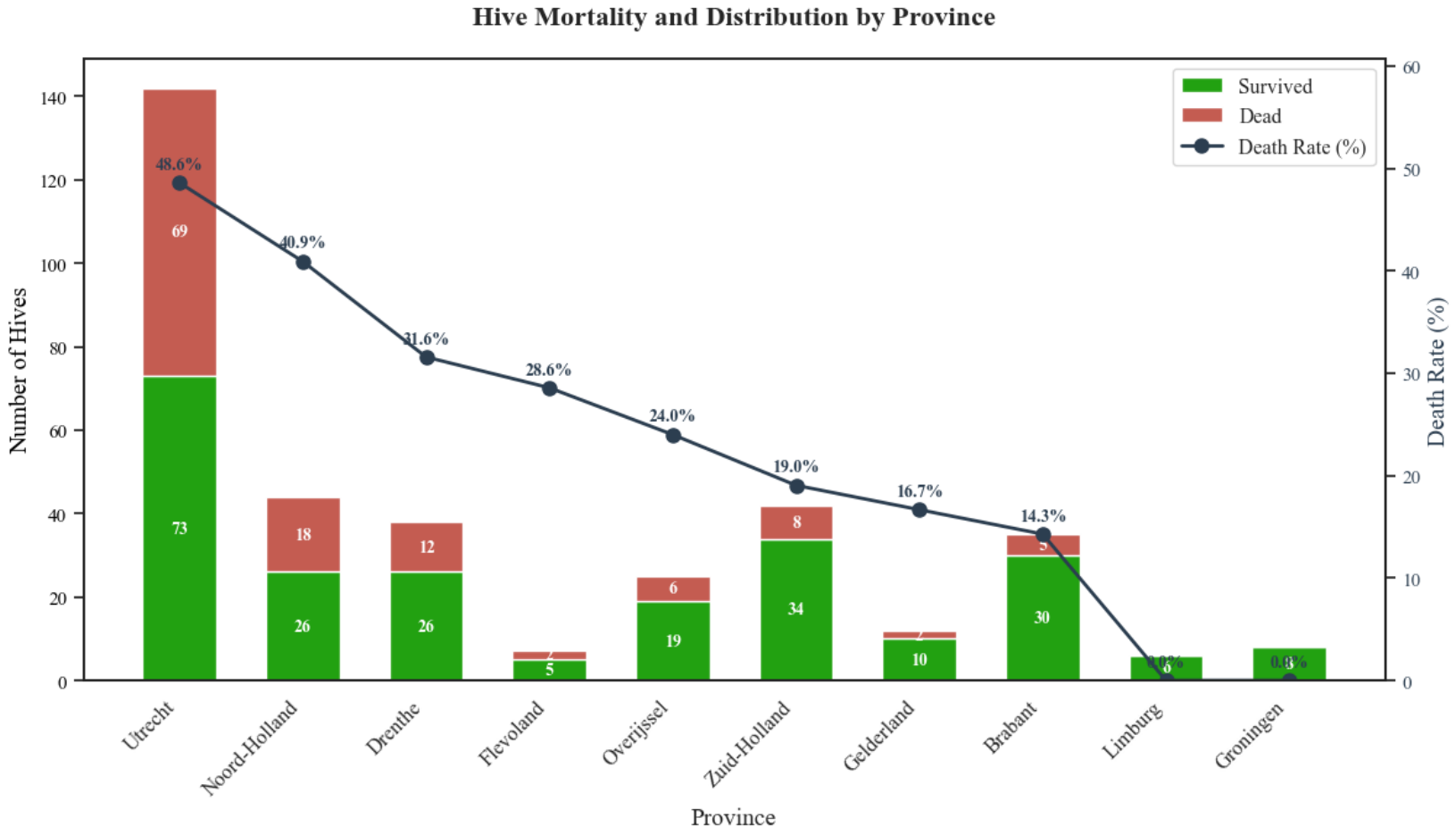


Geographical Distribution of the Hives

- We collected in total 63 beekeepers' answer of the GPS location of their hives, of which 6 of them have multiple hive locations. However, we didn't collect the detailed situation of each location, so these 6 results are omitted. In total there are 57 locations into the final analysis.
- These 57 locations covers 359 hives throughout 2024, where 20 of them are lost before the winter of 2024, 102 died during the winter of 2024, and 237 survived the winter of 2024.

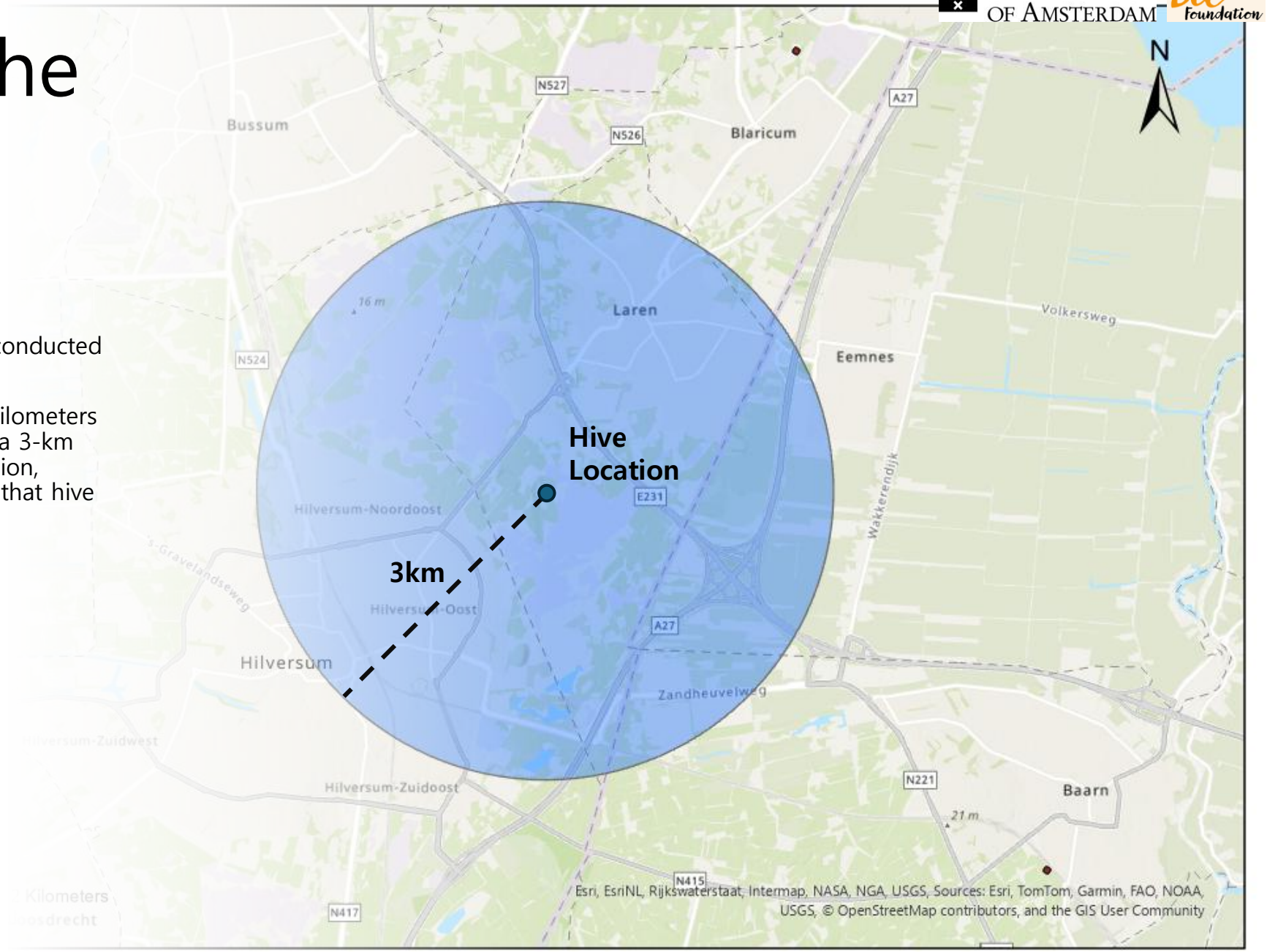
Province	Number of Beekeepers
Brabant	5
Drenthe	2
Flevoland	1
Gelderland	3
Groningen	1
Limburg	1
Noord-Holland	12
Overijssel	5
Utrecht	16
Zuid-Holland	11
Grand Total	**57**

Distribution by province of beekeepers who provided the GPS location of the hives.



Processing the GPS of the Hives

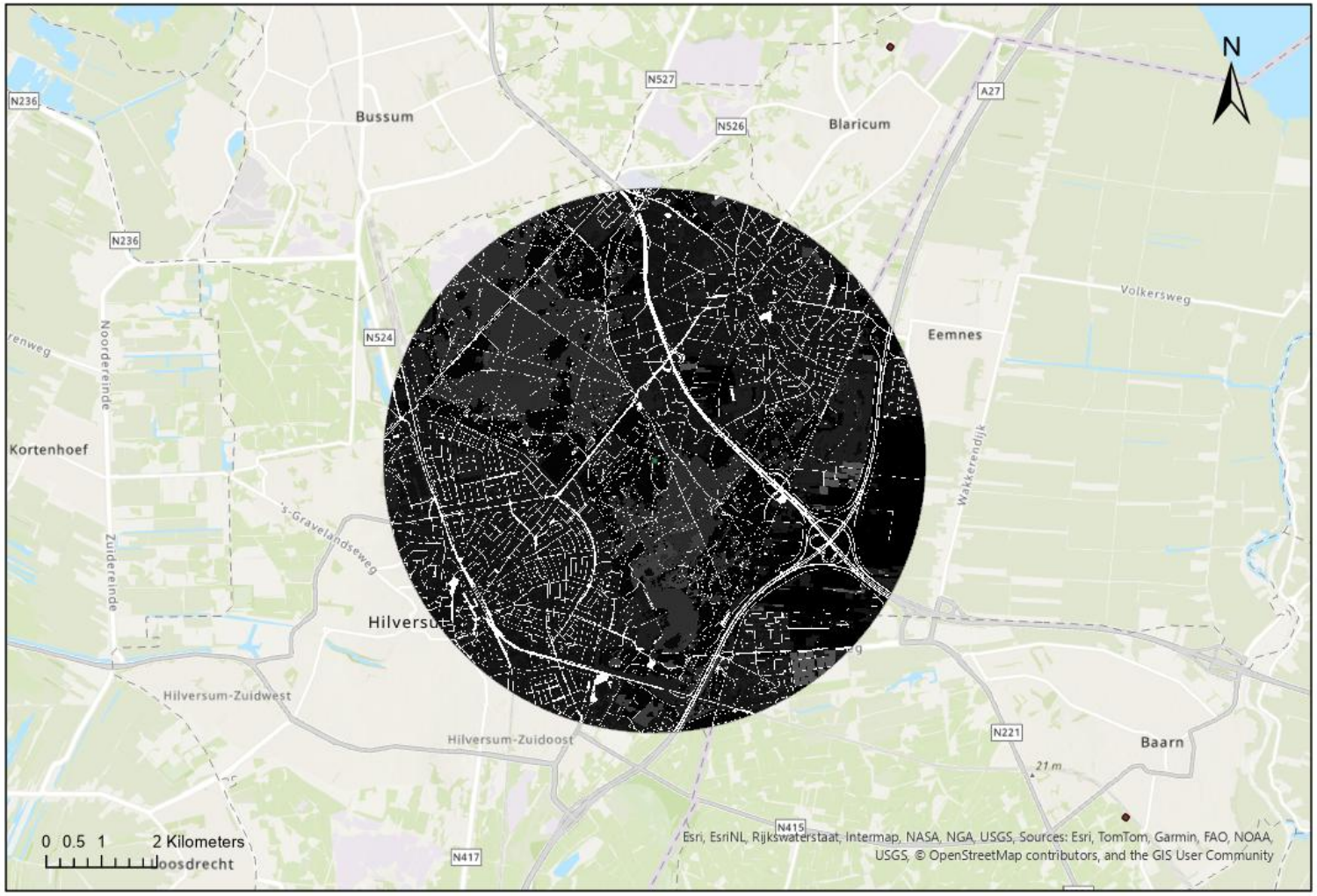
- With hive-level coordinates, we also conducted a spatial analysis.
- Honey bees can fly up to roughly 3 kilometers from their hive, therefore we created a 3-km radius buffer zone for each hive/location, representing the area that bees from that hive can potentially reach.



Using LGN data for Analyzing Land Use

- LGN: The Land Use Database of the Netherlands (Het Landelijk Grondgebruiksbestand Nederland - LGN) maps the entire country and provides the user with current and accurate information about land use in the Netherlands.
    - Size of land parcel: At a spatial resolution of 5*5m
- LGN2023 identifies 51 different types of land use.
    - *Het huidige bestand (LGN2023) geeft het landgebruik weer voor 51 klassen (verdeeld over landbouw, bos, water, natuur, stedelijk gebied) met een ruimtelijke resolutie van 5m.* https://lgn.nl/over-lgn
    - More in Appendix



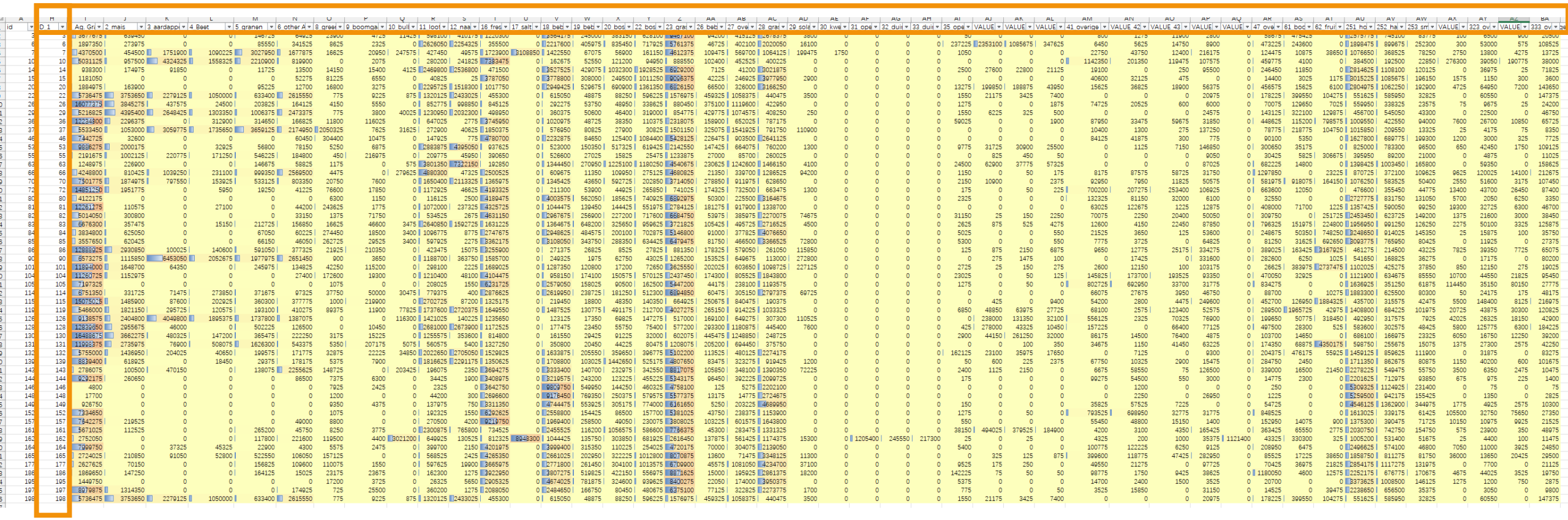


Using LGN data for Analyzing Land Use

- Different types of land use may involve different levels of pesticide use, which can influence the amount of pesticide residues that accumulate in honey.
- To explore this relationship, we projected the LGN land use layer to the 3-kilometer buffer zone. Within this area, we calculated the exact total area of different land types. This allowed us to determine exactly what kind of environment each colony was living in and its potential exposure risks to pesticides.

Using LGN data for Analyzing Land Use

Column: Land use type



Row: Hive Locations

Land Use of 3km Radius for Each Hive Location

Zooming in to a few specific locations…

- Criteria for selecting hives:

Zero death hives

- > 5 hives in from the same location
- Death rate: = 0%

- In total: 4 locations

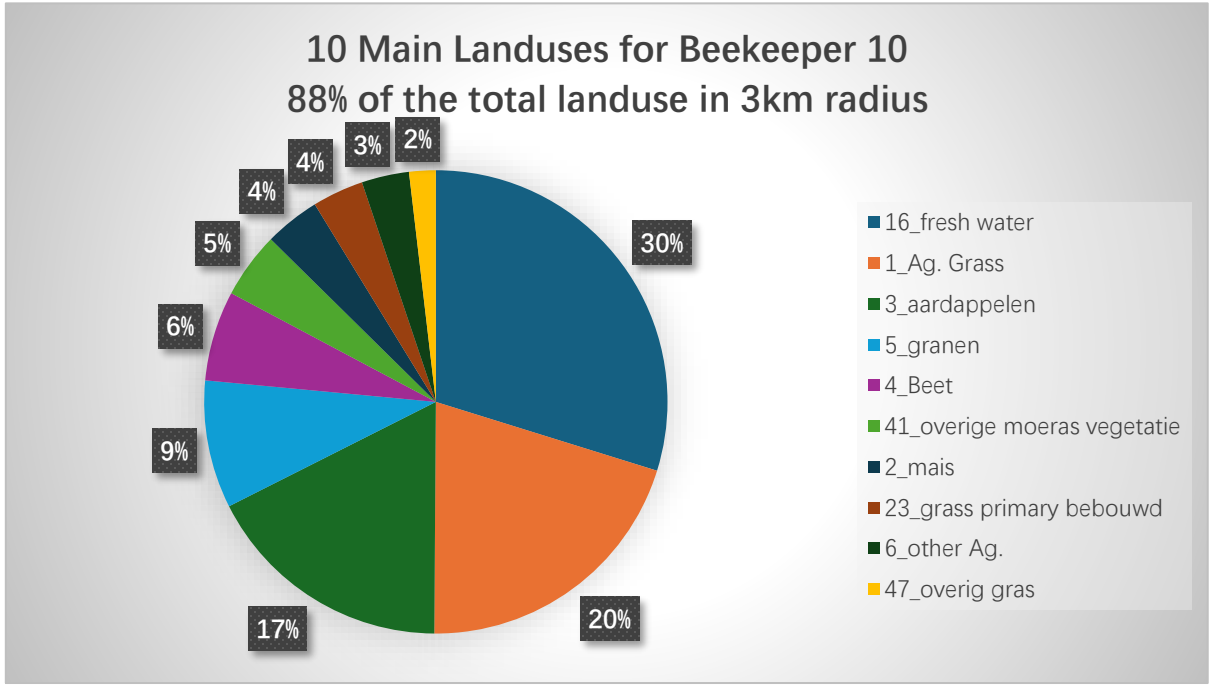
High death hives

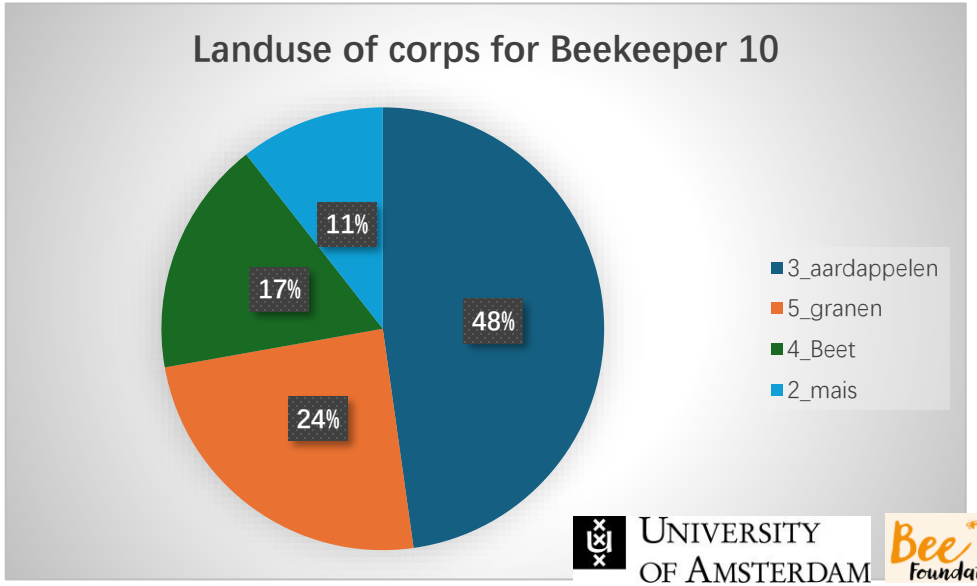
- > 10 hives in from the same location
- Death rate: >= 40%

- In total: 3 locations

Beekeeper id = 10
Province: Groningen
Survived winter: 20;
Died in Winter: 0;
Death Rate: 0%

The 3 km radius coverage has a large proportion of fresh water (27.4%), which is not that common compared with other locations.
The 4 kinds of crops takes up 31.68% of the 3 km radius, where potatoes takes up half of it.

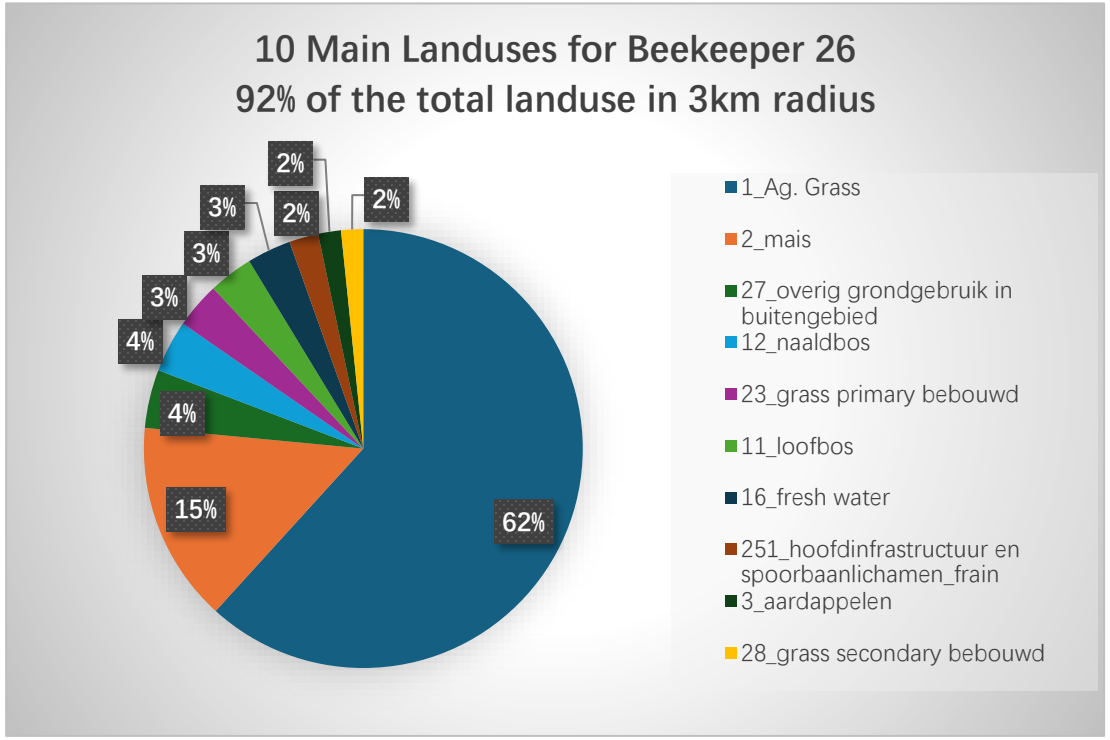




*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 26
Province: Overijssel
Survived winter: 7;
Died in Winter: 0;
Death Rate: 0%

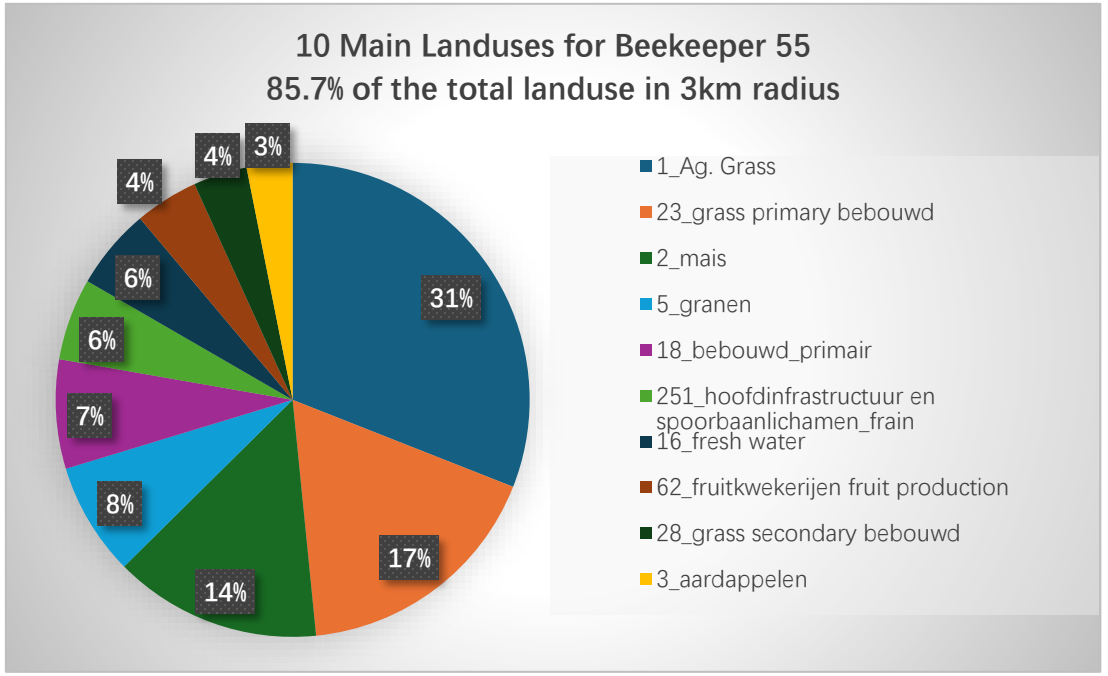
The 3 km radius covers a large area of agricultural grass (57.04%), followed by mais (13.8%).
There is only a very small proportion of the other 3 kinds of crops.

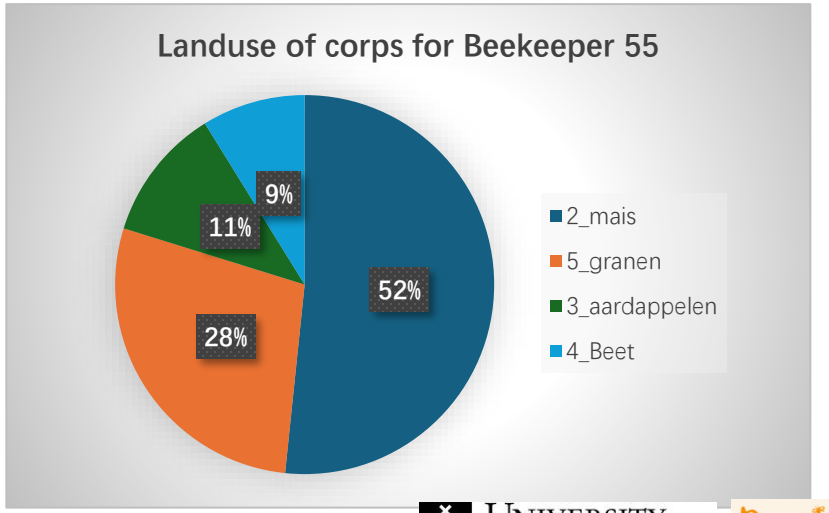


*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 55
Province: Limburg
Survived winter: 6;
Died in Winter: 0;
Death Rate: 0%

The 3 km radius coverage has a large proportion (51.42%) of agricultural land, including fruitkwekerijen.
The 4 kinds of crops takes up 23.50% of the 3km radius, where mais takes up half of it.

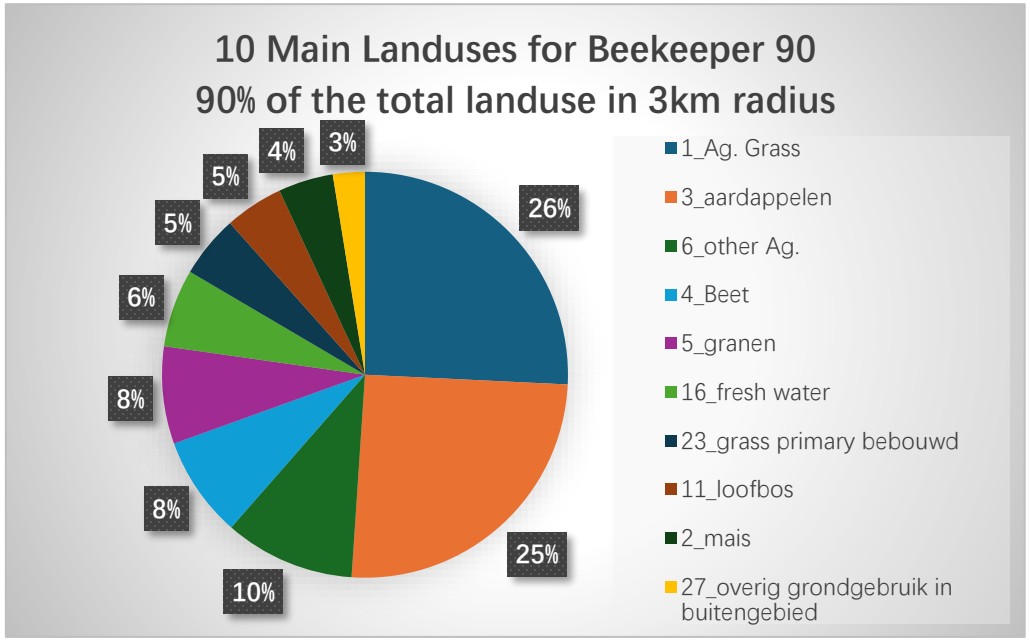


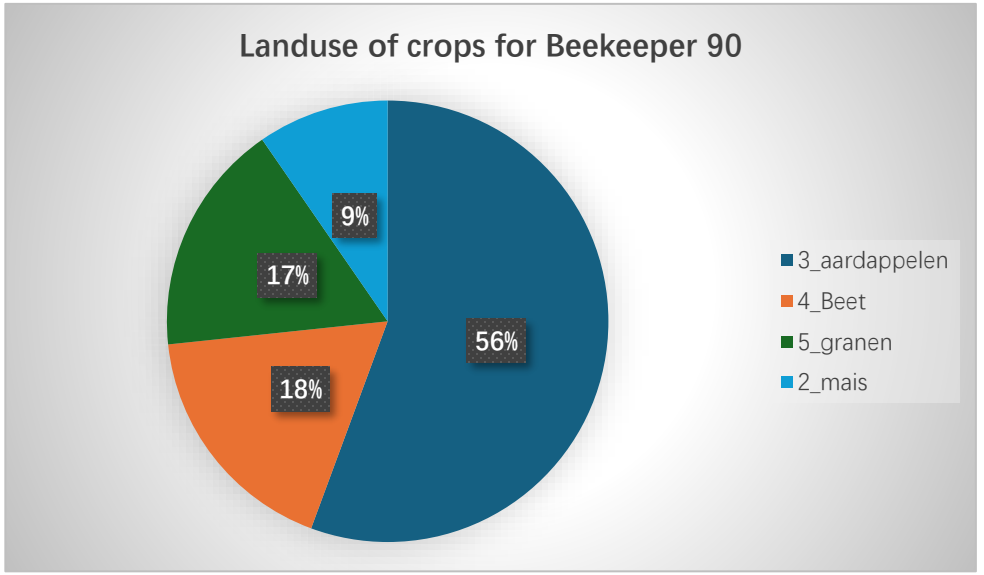


*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 90
Province: Groningen
Survived winter: 8;
Died in Winter: 0;
Death Rate: 0%

The 4 kinds of crops takes up 40.5% of the 3 km radius, where potatoes takes up more than half of it.

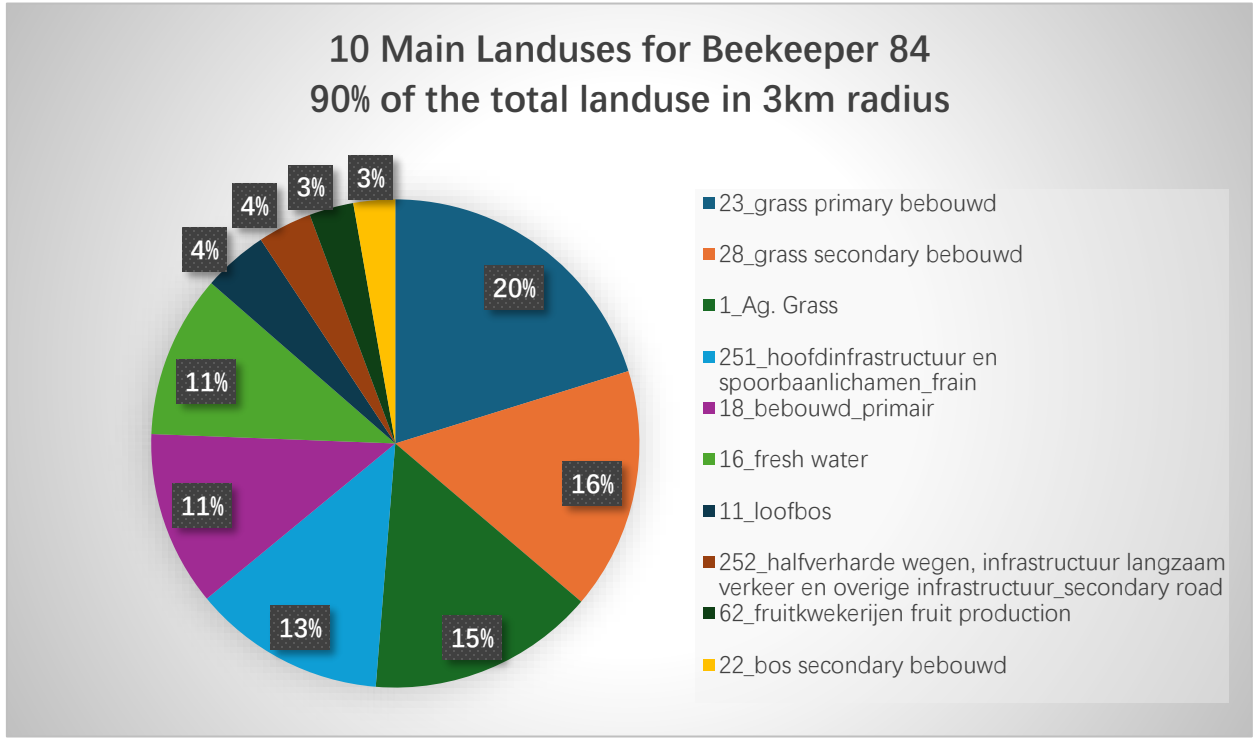




*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 84
Province: Utrecht
Survived winter: 3;
Died in Winter: 8;
Death Rate: 72.7%

The 3km radius covers mostly built areas.
There is no crops in the top 10 land uses.

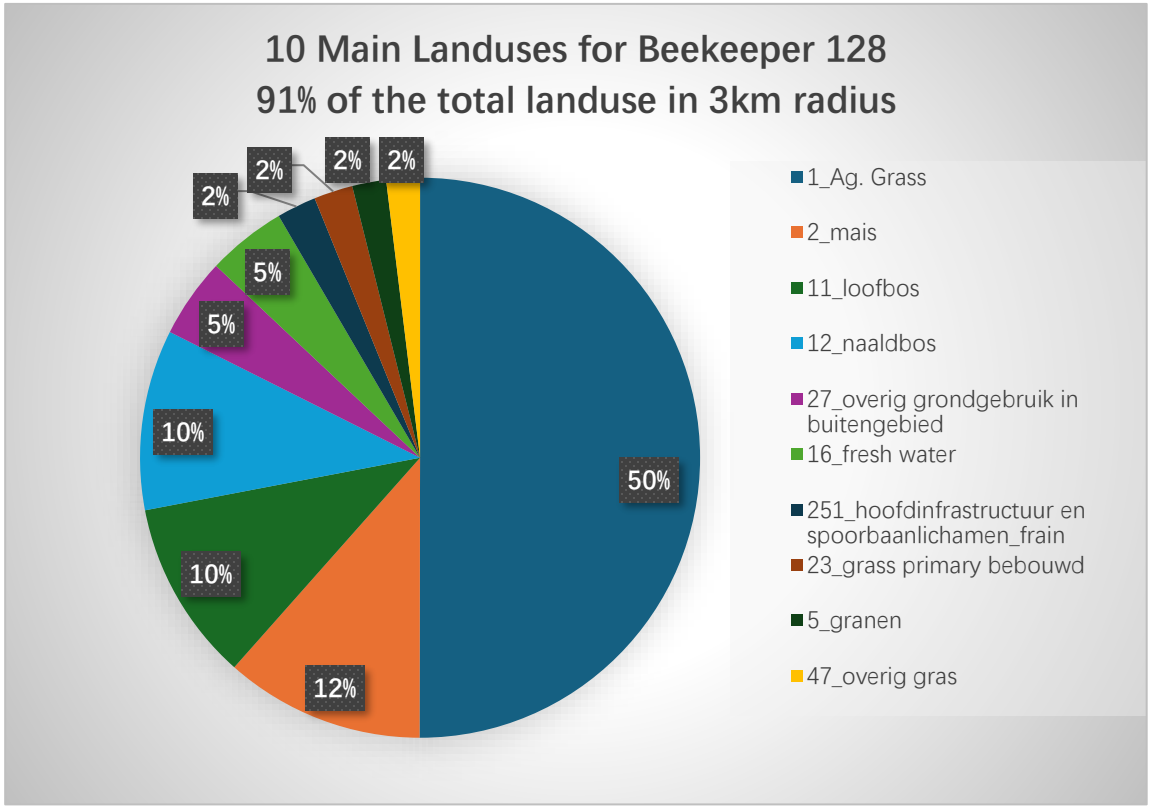


*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 128
Province: Overijssel
Survived winter: 6;
Died in Winter: 4;
Death Rate: 40%

The 3 km radius coverage has a large proportion of agricultural grass (45.5%), followed by mais (10.92%).
There is only a very small proportion of the other 3 kinds of crops.

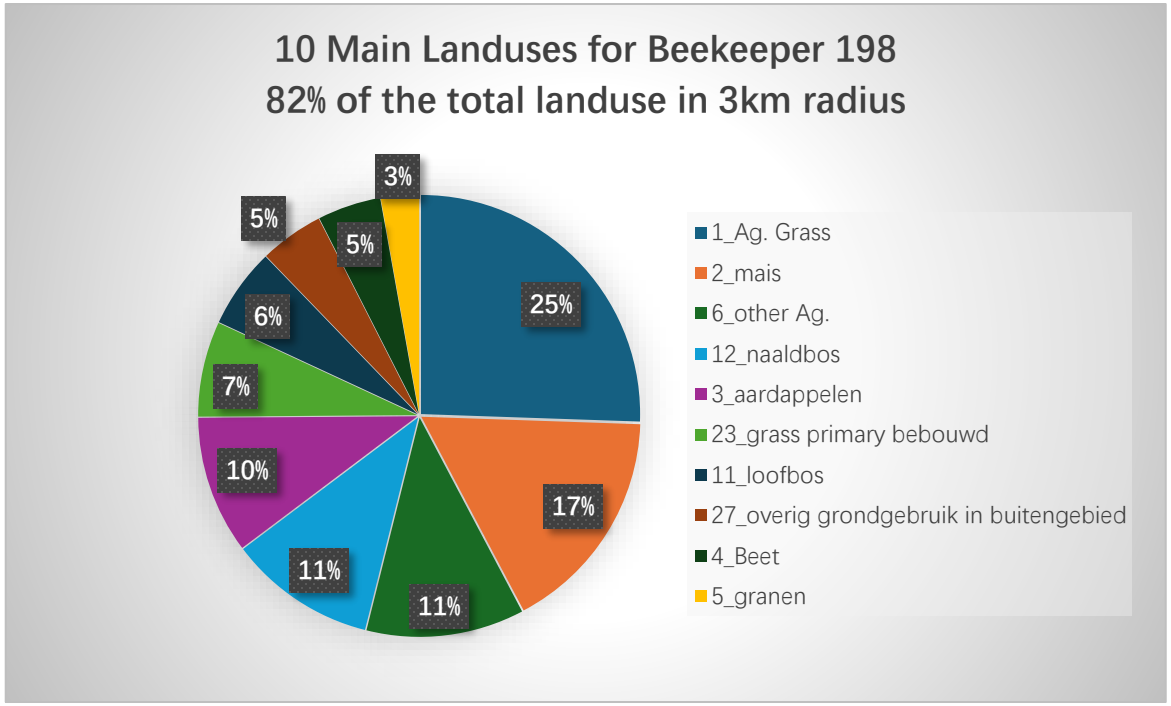
This land use pattern is similar with id = 26, but the death rate differs.

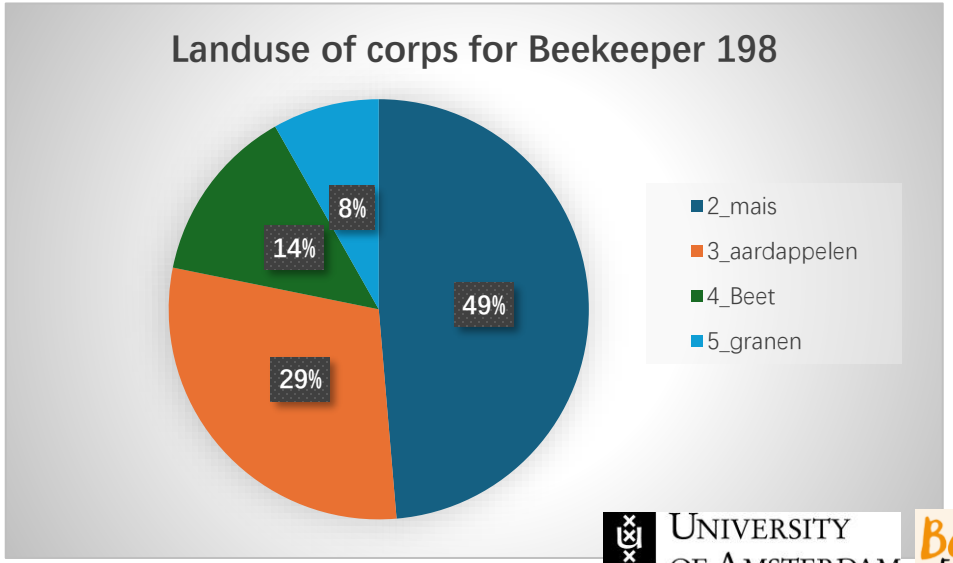


*To comply with data privacy requirements, the exact locations and detailed maps have been omitted from this report.

Beekeeper id = 198
Province: Brabant
Survived winter: 31;
Died in Winter: 49;
Death Rate: 69.25%

The 3 km radius covers a large area of agricultural land, where agricultural grass, the four kinds of crops, and other agricultural land takes up 20.5%, 28.7%, and 9.02%, respectively. Mais is the main crop in this area, with 13.92% of the 3km radius.





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GEE Regression Results

Dep. Variable:	Hive_Status	No. Observations:	359
Model:	GEE	No. clusters:	55
Method:	Generalized Estimating Equations	Min. cluster size:	1
		Max. cluster size:	80
Family:	Binomial	Mean cluster size:	6.5

*Land use type are calculated in percentage, instead of actual size.

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.3409	2.377	-0.143	0.886	-5.001	4.319
High_Risk_Ag_pct	-0.0295	0.024	-1.234	0.217	-0.076	0.017
Orchards_Greenhouses_pct	-0.0316	0.094	-0.334	0.738	-0.217	0.154
Grassland_pct	-0.0215	0.035	-0.616	0.538	-0.090	0.047
Nature_Seminatural_pct	0.0353	0.030	1.179	0.239	-0.023	0.094

Skew:	0.6430	Kurtosis:	-1.4201
Centered skew:	0.2774	Centered kurtosis:	-0.6949

Baseline Land Use:
Urban Infrastructure

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	3.1848	1.549	2.056	0.040	0.148	6.222
Urban_Infrastructure_pct	-0.0353	0.030	-1.179	0.239	-0.094	0.023
Orchards_Greenhouses_pct	-0.0668	0.092	-0.729	0.466	-0.247	0.113
Grassland_pct	-0.0567	0.025	-2.312	0.021	-0.105	-0.009
High_Risk_Ag_pct	-0.0647	0.025	-2.586	0.010	-0.114	-0.016

Skew:	0.6430	Kurtosis:	-1.4201
Centered skew:	0.2774	Centered kurtosis:	-0.6949

Baseline Land Use:
Nature and Semi-nature

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-2.4864	1.340	-1.855	0.064	-5.113	0.140
High_Risk_Ag_pct	-0.0080	0.022	-0.372	0.710	-0.050	0.034
Orchards_Greenhouses_pct	-0.0101	0.102	-0.099	0.921	-0.209	0.189
Urban_Infrastructure_pct	0.0215	0.035	0.616	0.538	-0.047	0.090
Nature_Seminatural_pct	0.0567	0.025	2.312	0.021	0.009	0.105

Skew:	0.6430	Kurtosis:	-1.4201
Centered skew:	0.2774	Centered kurtosis:	-0.6949

Baseline Land Use:
Grassland

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-3.2871	1.446	-2.274	0.023	-6.121	-0.453
Urban_Infrastructure_pct	0.0295	0.024	1.234	0.217	-0.017	0.076
Orchards_Greenhouses_pct	-0.0021	0.099	-0.021	0.983	-0.196	0.192
Grassland_pct	0.0080	0.022	0.372	0.710	-0.034	0.050
Nature_Seminatural_pct	0.0647	0.025	2.586	0.010	0.016	0.114

Skew:	0.6430	Kurtosis:	-1.4201
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Baseline Land Use:
High Risk Agriculture

- Our GEE analysis consistently reveals that proximity to nature and semi-natural areas is associated with higher winter mortality, compared to agricultural or grassland landscapes.
- This requires further analysis to explain the underlying mechanisms.

Next Step?

- Land use analysis:
  - What is a better way to aggregate land use?
  - What are the specific land that deserve a higher focus?
- How to analyze the effect of pesticide on bee mortality? What is an operational way to link land use with the toxicity of the surrounding environment?
  - Which data source to use and where to get them?
    - Surface Water?
    - (Legal, estimated) Pesticide residue in crops and grass?
    - (Actual) Pesticide residue?

Appendix

Reference

- Vervaardiging en nauwkeurigheid van het LGN3-grondgebruiksbestand.
- Wit, A.J.W. de, Heijden, Th.G.C. van der en H.A.M. Thunnissen, 1999. Wageningen, DLO-Staring Centrum, Rapport 663.
- Landelijk Grondgebruiksbestand Nederland (LGN5). Vervaardiging, nauwkeurigheid en gebruik.
- Hazeu, G.W., 2005. Wageningen, Alterra. Alterra-report 1213, 92p., 18 figs., 11 tables and 11 refs. DOI: https://doi.org/10.18174/17654
- Landelijk Grondgebruiksbestand Nederland versie 6 (LGN6). Vervaardiging, nauwkeurigheid en gebruik.
- Hazeu, G.W., Schuiling, C., Dorland, G.J., Oldengarm, J. & H.A. Gijsbertse, 2010. Wageningen, Alterra. Alterra-rapport 2012, 132p., 20 figs., 9 tables and 15 refs. DOI: https://doi.org/10.18174/137531

- Hazeu, G., R. Schuiling, D. Thomas, M. Vittek, M. Storm en J.D. Bulens, 2023. Wageningen, Wageningen Environmental Research, Rapport 3235. 100 blz.; 10 fig.; 8 tab.; 16 ref. https://lgn.nl/reports/WENR-rapport%203235_Totaal_LR.pdf#page=59.11'
- Ctgb. Guidelines for maximum residue levels in honey. https://english.ctgb.nl/latest/news/2019/05/06/guidelines-for-maximum-residue-levels-in-honey
- Technical guidelines for determining the magnitude of pesticide residues in honey and setting Maximum Residue Levels in honey. https://food.ec.europa.eu/document/download/06bcc245-57b5-47ff-b31b-25c05a8485cf_en?filename=pesticides_mrl_guidelines_honey.pdf

Geographical Distribution of the Hives

- We collected in total 63 beekeepers' answer of the GPS location of their hives, of which 6 of them have multiple hive locations. However, we didn't collect the detailed situation of each location, so these 6 results are omitted. In total there are 57 locations into the final analysis.
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Zuid-Holland	11
Grand Total	**57**

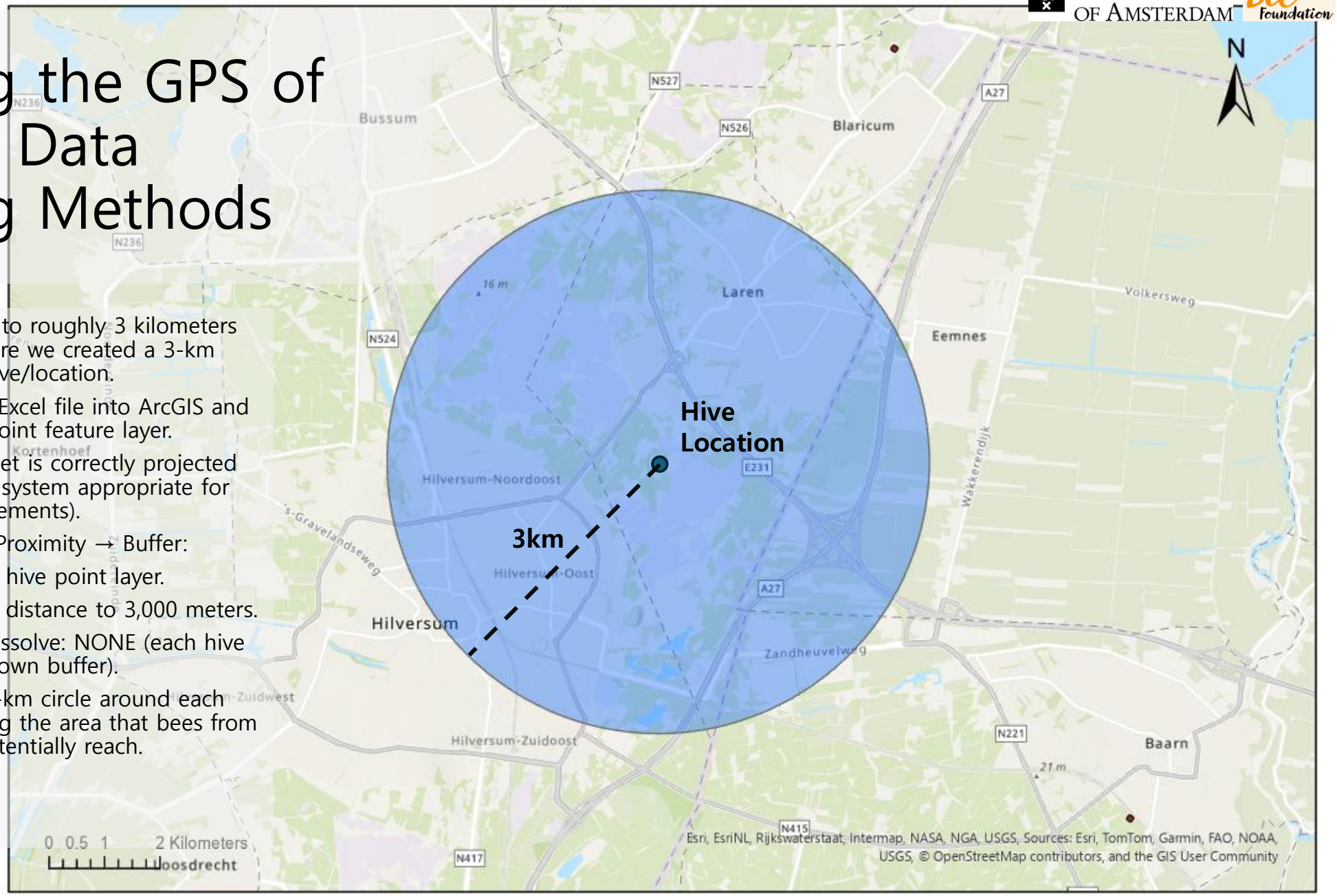
Distribution by province of beekeepers who provided the GPS location of the hives.

Province	Died During Winter 2024	Survived Winter 2024	Grand Total
Brabant	4	30	**34**
Drenthe	2	26	**28**
Flevoland	1	5	**6**
Gelderland	2	10	**12**
Groningen	0	8	**8**
Limburg	0	6	**6**
Noord-Holland	12	26	**38**
Overijssel	6	19	**25**
Utrecht	68	73	**141**
Zuid-Holland	7	34	**41**
Grand Total	**102**	**237**	**339**

Distribution by province of hives with a GPS location.

Processing the GPS of the Hives: Data Processing Methods

- Honey bees can fly up to roughly 3 kilometers from their hive, therefore we created a 3-km buffer zone for each hive/location.
  - Import the hive Excel file into ArcGIS and convert it to a point feature layer.
  - Ensure the dataset is correctly projected (e.g., coordinate system appropriate for distance measurements).
  - Use Analysis → Proximity → Buffer:
    - Select the hive point layer.
    - Set buffer distance to 3,000 meters.
    - Choose dissolve: NONE (each hive needs its own buffer).
  - This created a 3-km circle around each hive, representing the area that bees from that hive can potentially reach.



Analysis on Land Use

- As there are more than 40 land use variables and only 57 different locations, including all the land use into the model has a risk of overfitting. Therefore, we aggregated land use into 5 major groups based on their ecological relevance. The cumulative area of land use types within each group was then calculated for each buffer zone.

Code	Hoofdklasse	Subgroep	Klasse
2	High risk agriculture		Mais
3			Aardappelen
4			Bieten
5			Granen
6			Overige landbouwgewassen
10			Bloembollen
61			Boomkwekerijen
8	Orchards and Greenhouses		Glastuinbouw
9			Boomgaarden
62			Fruitkwekerijen
18	Urban Infrastructure	Bebouwd gebied	Bebouwing in primair bebouwd gebied
19			Bebouwing in secundair bebouwd gebied
26			Bebouwing in buitengebied
27			Overig grondgebruik in buitengebied
29			Zonneparken
251			Hoofdinfrastructuur en spoorbaanlichamen
252			Halfverharde wegen, infrastructuur langzaam verkeer en overige infrastructuur
253			Smalle wegen
1	Grassland and managed forests		Agrarisch gras
23			Gras in primair bebouwd gebied
28			Gras in secundair bebouwd gebied
46			Gras in het kustgebied
47			Overig gras
11	Nature and Seminature	Bos	Loofbos
12			Naaldbos
20			Bos in primair bebouwd gebied
22			Bos in secundair bebouwd gebied
45		Gras	Natuurlijk beheerde agrarische graslanden
16		Water	Zoet water
17			Zout water
30		Kustgebied	Kwelders
31			Open zand in kustgebied
32			Duinen met lage vegetatie (<1m)
33			Duinen met hoge vegetatie (>1m)
35		Heidegebied	Open stuifzand en/of rivierzand
36			Heide
37			Matig vergraste heide
38			Sterk vergraste heide
41		Moeras	Overige moerasvegetatie
42			Rietvegetatie
43			Bos in moerasgebied
322			Struikvegetatie in moerasgebied (laag)
323			Overige struikvegetatie (laag)
332			Struikvegetatie in moerasgebied (hoog)
333			Overige struikvegetatie (hoog)

Land Use Code Definition

- In general
  - The land-use code is updated each time a new LGN report is made, though most categories remain the same. Since this analysis uses LGN 2023 data, we follow the LGN 2023 coding system. All code versions are available under the 'Documentatie' section.
  - Because some land-use categories are not self-explanatory, the following pages include short descriptions of each.
  - Zoals de naam van hetbestand al aangeeft betreft het een grondgebruiksbestand. Dit betekent dat behalve het bedekkingstype van het betreffende gebied ook het gebruikof de functie van invloed is op de indeling in klassen. / As the name of the file indicates, it is a land use file. This means that in addition to the land cover type of the area in question, the use or function also influences the classification. (C-DLO Rapport 663O, p.35)
- Agrarisch gras / Agricultural grass:
  - Definition haven't changed since LGN 3+
  - Grasland binnen het stratum agrarisch gebied. Dit betreft voor het overgrote deel **grasland dat gebruikt wordt voor agrarische productie**, maar voor een deel betreft het ook erven van **boerderijen en bedrijven, gras op dijken, wegbermen en andere met gras bedekte oppervlakken**. / Grassland within the agricultural area stratum. This mainly concerns **grassland used for agricultural production**, but it also includes **farmyards and businesses, grass on dikes, roadside verges, and other grass-covered surfaces**. (Hazeu, G.W., 2005, p.65)
- Boomkwekerijen en Fruitkwekerijen
  - Definition haven't changed since LGN 6.
  - Boomkwekerijen zijn **percelen in gebruik voor het opkweken van bomen en struiken** waarbij hoogte van de aanplant niet van belang is. / Tree nurseries are **plots used for growing trees and shrubs** where planting height is not a factor. (Alterra-rapport 2012, p.72
  - Fruitkwekerijen zijn **percelen met laagstammige fruitbomen** zonder onderscheid naar type vrucht. / Fruit nurseries are **plots with low-stemmed fruit trees**, with no distinction made based on fruit type. (Alterra-rapport 2012, p.72)

Land Use Code Definition (Continued)

- Gras in primair bebouwd gebied
  - Definition haven't changed since LGN 6.
  - Overwegend met gras bedekte gebieden met een stedelijke functie. Het betreft **o.a. bouwterreinen, parken en plantsoenen, sportterreinen** die behoren tot het primair bebouwd gebied. Het primair bebouwd gebied is het bebouwd gebied zoals gedefinieerd door DGR van het ministerie van VROM. Het is aangevuld met missende vlakken van de BBG2003-klassen woongebied, detailhandel en horeca, bedrijfsterreinen en bouwterreinen. / Predominantly grass-covered areas with an urban function. This includes **building sites, parks and public gardens, and sports fields** that belong to the primary built-up area. The primary built-up area is the built-up area as defined by the DGR of the Ministry of Housing, Spatial Planning and the Environment (VROM). It has been supplemented with missing areas from the BBG2003 classes residential, retail and hospitality, business parks, and building sites. natuur (Alterra-rapport 2012, p.74)
- Gras in secundair bebouwd gebied
  - Definition haven't changed since LGN 6.
  - **Golf court is under this section.**
  - Overwegend met gras bedekte gebieden met een stedelijke functie. Het betreft met namen **. sportterreinen, recreatiegebieden, golfterreinen, stortplaatsen maar ook vliegvelden**. Het semi/secundair bebouwd gebied is de niet in het primair bebouwd gebied **opgenomen vliegvelden, openbare voorzieningen, sociaal-culturele voorzieningen, stortplaatsen, wrakkenopslag-, begraaf- en delfstofwinplaatsen** uit het BBG2003-bestand. Verder bevat het de **parken en plantsoenen, sportterreinen, volkstuinen, dagrecreatieve terreinen en verblijfsrecreatie** uit het BBG2003-bestand. / Predominantly grass-covered areas with an urban function. This includes, for example**, sports fields, recreation areas, golf courses, landfills, and airports**. The semi-/secondary built-up area comprises **the airports, public facilities, socio-cultural facilities, landfills, wreck storage facilities, burial sites, and mineral extraction sites** from the BBG2003 database that are not included in the primary built-up area. It also includes **the parks and public gardens, sports fields, allotments, day recreation areas, and residential recreation facilities** from the BBG2003 database. (Alterra-rapport 2012, p.74)

Land Use Code Definition (Continued)

- Bebouwing in primair bebouwd gebied
  - Definition haven't changed since LGN 6.
  - Bebouwing vallend binnen de LGN-klasse 23 (gras in primair bebouwd gebied). / Buildings falling within LGN class 23 (grass in primary built-up areas). (Alterra-rapport 2012, p.73)
- Bebouwing in secundair bebouwd gebied
  - Definition haven't changed since LGN 6.
  - Bebouwing vallend binnen de LGN-klasse 28 (gras in secundair bebouwd gebied). De bebouwing heeft geen agrarische functie. Hierbij moet worden gedacht **aan gebouwen op vliegvelden, bungalowparken en campings, gebouwen op militaire terreinen, gebouwen van de elektriciteitsvoorziening, waterzuiveringsinstallaties, gebouwen op sportterreinen, in recreatieparken en in parken en plantsoenen.** / Buildings falling within LGN class 28, grass in secondary built-up areas. These buildings do not have an agricultural function. This includes **buildings at airports, bungalow parks and campsites, buildings on military sites, buildings for electricity supply, water treatment plants, buildings on sports fields, in recreation parks, and in parks and public gardens**. (Alterra-rapport 2012, p.73)
- Bebouwing in buitengebied
  - Definition haven't changed since LGN 6.
  - Bebouwing binnen het stratum agrarisch gebied, bos en natuur. / Buildings within the stratum agricultural area, forest and nature (Alterra-rapport 2012, p.74).
  - De bebouwing in het buitengebied is gedefinieerd als zijnde de **bebouwing vallende binnen de klassen secundair bebouwd gebied** (LGN klasse 28), **bos in secundair bebouwd gebied** (LGN klasse 22) en **kale grond in bebouwd buitengebied** (LGN klasse 24). / Buildings in rural areas are defined as **buildings falling within the classes secondary built-up area** (LGN class 28), **forest in secondary built-up area** (LGN class 22), and **bare ground in rural built-up areas** (LGN class 24). (Alterra-rapport 2012, p.73)

Reference for land use categorization

Opbouw legenda Landelijk Grondgebruiksbestand Nederland versie 6.

Code	Hoofdklasse	Subgroep	Klasse
1	Agrarische gebied		Agrarisch gras
2			Mais
3			Aardappelen
4			Bieten
5			Granen
6			Overige landbouwgewassen
61			Boomkwekerijen
62			Fruitkwekerijen
8			Glastuinbouw
9			Boomgaard
10			Bollen
26			Bebouwing in buitengebied
11	Bos		Loofbos
12			Naaldbos
16	Water		Zoet water
17			Zout water
18	Bebouwd gebied		Bebouwing in primair bebouwd gebied
19			Bebouwing in secundair bebouwd gebied
20			Bos in primair bebouwd gebied
22			Bos in secundair bebouwd gebied
23			Gras in primair bebouwd gebied
24			Kale grond in bebouwd buitengebied
28			Gras in secundair bebouwd gebied
25	Infrastructuur		Hoofdwegen & spoorwegen
30	Natuur	Kustgebied	Kwelders
31			Open zand in kustgebied
32			Duinen met lage vegetatie (<1m)
33			Duinen met hoge vegetatie (>1m)
34			Duinheide
35		Heidegebied	Open stuifzand en/of rivierzand
36			Heide
37			Matig vergraste heide
38			Sterk vergraste heide
39		Hoogveen	Hoogveen
40			Bos in hoogveengebied
41		Moeras	Overige moerasvegetatie
42			Rietvegetatie
43			Bos in moerasgebied
45			Natuurgraslanden

(Alterra-rapport 2012, p.17)

Certain urban/rural land use for specific functions

- **Golfterrein – 2 – secondary built area**

- De tabel geeft voor de Functionele Gebieden uit TOP10NL aan welke klasse van het stedelijk masker zeworden toegekend. Functionele gebieden die de code 1 krijgen, behoren tot het primair stedelijke gebied. Gebieden met code 2 behoren tot het secundair stedelijke gebied en gebieden met code 0 behoren tot hetbuitengebied (Wageningen Environmental Research Rapport 3235, p.74) https://lgn.nl/reports/WENR-rapport%203235_Totaal_LR.pdf

- Acelepryn Controversy: The CTGB has authorized Acelepryn (Chlorantraniliprole) for use against leatherjackets (emelten) and grubs (engerlingen) on golf courses and sports fields.(https://www.syngenta.nl/news/gewasbescherming/ctgb-besluit-tot-toelating-van-insecticide-acelepryn-voor-sportvelden-en) This authorization exists despite the BSNC (industry association) advising against its use to adhere to Green Deal goals.(https://www.onkruidvergaat.nl/bsnc-adviseert-beheerders-van-sportvelden-en-golfbanen-om-acelepryn-niet-te-gebruiken-tegen-engerlingen-en-emelten/)
- Impact: Chlorantraniliprole is a ryanodine receptor modulator. It causes muscle paralysis in insects. While it has a very low acute toxicity profile for adult bees, it affects larval development. If golf courses (often perceived as green spaces) are treated with Acelepryn, and bees collect water (guttation drops) or forage on clover in the rough, the chemical can enter the hive, potentially affecting brood rearing in the critical late-summer window.42

Functioneel gebied	Code
attractiepark	2
bedrijventerrein	1
begraafplaats	2
botanische tuin	2
bungalowpark	2
camping, kampeerterrein	2
campus	2
caravanpark	2
circuit	2
crossbaan	2
dierentuin, safaripark	2
eendenkooi	0
emplacement	2
erebegraafplaats	2
gaswinning	2
gebied voor radioastronomie	2
gebouwencomplex	1
golfterrein	2
grafheuvel	0
grindwinning	2
groeve	2
haven	1
heemtuin	2
helikopterlandingsterrein	2
ijsbaan	2
jachthaven	2
kartingbaan	2
kassengebied	0
kazerne, legerplaats	2
landgoed	0
mijn	2

Using LGN data for Analyzing Land Use: Data Processing Methods

- Different types of land use may involve different levels of pesticide application, which can influence the amount of pesticide residues that accumulate in honey. To explore this relationship, we use a 3-km buffer around each hive to identify the surrounding land-use composition. This spatial characterization serves as the basis for further analysis linking land-use patterns with potential pesticide exposure.
- In order to achieve this goal, each buffer zone needed to be intersected with land-use data.
  - Import a land-use classification layer from LGN (raster).
  - Convert raster categories to polygons
  - Use Spatial Analyst → Extraction → Extract by Mask, intersect each hive's 3-km buffer with the land-use layer, we get a new raster layer containing only land-use areas that fall inside each buffer.
- Next, we calculated area by land-use type.
  - Use Spatial Analyst → Zonal → Tabulate Area, Run Tabulate Area with beekeeper_id as zones and landuse buffer raster as classes, we get a new dataset containing only land-use areas per class per buffer.
  - This gives total area per land-use type for every hive's 3-km buffer zone
- The final step is to exporting results into an Excel file after calculating land-use composition.
  - Use Table → Export Table to save the summary statistics as a CSV/Excel file.
- The final product is an Excel file containing:
  - *beekeeper_id* (which corresponds to the GPS location specified by the beekeeper)
  - *total area of each land-use type within the 3-km buffer*